

# Kidney-on-a-Chip Microphysiological Systems: Physiological Foundations, Design Landscape, Critical Gaps, and Future Directions

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## 1. Microphysiological Systems (MPS) and Organ-on-a-Chip: Definitions and Scope

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Microphysiological systems (MPS) are in vitro platforms composed of living human or animal cells maintained within microengineered environments that replicate the biochemical, mechanical, and structural cues of native organs. As defined by the U.S. National Toxicology Program Interagency Center for the Evaluation of Alternative Toxicological Methods (NICEATM), an MPS is an in vitro platform that provides biochemical, electrical, or mechanical responses to model-specific properties that define organ or tissue function. MPS encompasses organ-on-a-chip (OoC) devices, organoids, 3D co-culture constructs with extracellular matrix, and 3D bioprinted tissue constructs.

An organ-on-a-chip (OoC) is specifically a multi-channel, three-dimensional microfluidic cell-culture device, which is an integrated circuit for biology that simulates the activities, mechanics, and physiological responses of an entire organ or organ system. First established by Huh et al. (2010) with the landmark lung-on-a-chip, OoC systems introduced the concept of living microarchitecture: cells are seeded on optically transparent, gas-permeable membranes (typically polydimethylsiloxane, PDMS) within continuously perfused microchannels that impose physiological fluid shear stress, stretch, and concentration gradients. The convergence of lab-on-chip (LOC) technology with cell biology creates experimental platforms that outperform both static 2D culture and animal models in predicting human drug responses and disease phenotypes.

Key functional attributes of MPS/OoC include:

- (i) Continuous microfluidic perfusion replicating vascular or luminal flow
- (ii) 3D tissue interfaces allowing co-culture of epithelial, endothelial, and stromal compartments
- (iii) Mechanotransduction via programmable cyclic stretch or shear stress
- (iv) Real-time optical and electrochemical sensing integrated on-chip
- (v) Scalability toward multi-organ body-on-chip configurations linking gut, liver, kidney, and lung in pharmacokinetic circuits.

## 2. New Approach Methodologies (NAMs): Definition and Regulatory Context

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New Approach Methodologies (NAMs) is a term formally defined by the Interagency Coordinating Committee on the Validation of Alternative Methods (ICCVAM) as any technology, methodology, approach, or combination thereof that can provide information on chemical hazard and risk assessment that replaces, reduces, or refines the use of animals. The concept is grounded in the 3Rs framework: Replacement, Reduction, and Refinement, originally articulated by Russell and Burch (1959), and has gained extraordinary regulatory momentum following the U.S. FDA Modernization Act 2.0 (2022) and the FDA Roadmap to Reducing Animal Testing in Preclinical Safety Studies (2024–2025).

NAMs encompass a broad, heterogeneous suite of technologies:

- In vitro complex models: organ-on-a-chip, organoids, 3D spheroids, air-liquid interface cultures
- In silico and computational tools: physiologically-based pharmacokinetic (PBPK) models, quantitative structure-activity relationship (QSAR) models, AI/machine learning toxicity prediction
- In chemico assays: reactive oxygen species detection, direct peptide reactivity assay (DPRA) for skin sensitization
- High-throughput and high-content screening platforms: automated imaging, multi-well microfluidic arrays
- Omics-integrated approaches: transcriptomics, proteomics, metabolomics profiling from MPS

As of March 2026, the FDA's Center for Drug Evaluation and Research (CDER) released draft guidance establishing four validation pillars for NAMs: (1) context of use, (2) human biological relevance, (3) technical characterization, and (4) fit-for-purpose regulatory utility. Microphysiological systems, especially kidney-on-chip models, are regarded as among the highest-fidelity NAMs, as they simultaneously incorporate human primary cells, physiological fluid dynamics, and real-time functional readouts that static in vitro systems fundamentally cannot replicate.

### 3. Key Physiological Parameters of the Kidney

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The kidney is one of the most functionally complex organs, processing approximately 180 L of plasma filtrate daily and maintaining systemic homeostasis through precise regulation of fluid volume, electrolyte balance, acid-base chemistry, and xenobiotic excretion. For accurate OoC modeling, four cardinal physiological parameters must be recapitulated:

#### 3.1 Glomerular Filtration

The glomerulus is a specialized capillary tuft enclosed within Bowman's capsule that performs primary plasma ultrafiltration. The glomerular filtration barrier (GFB) is a tri-layered structure comprising-

- (i) Fenestrated glomerular endothelial cells (GEnC) with their negatively charged glycocalyx.
- (ii) The glomerular basement membrane (GBM), a laminin- and collagen IV-rich extracellular matrix.
- (iii) Visceral epithelial cells (podocytes) with interdigitating foot processes bridged by nephrin-containing slit diaphragms.

This barrier enforces size selectivity (molecular weight cut-off  $\sim 70$  kDa for globular proteins) and charge selectivity (exclusion of anionic albumin). The glomerular filtration rate (GFR) in healthy adults is  $\sim 125$  mL/min and serves as the primary clinical index of renal function. On-chip, GFB integrity is assessed by albumin retention, inulin sieving coefficients, and dextran size-exclusion assays.

#### 3.2 Tubular Reabsorption and Active Secretion

The renal proximal tubule (PT) reabsorbs  $\sim 65$ – $70\%$  of filtered sodium, water, glucose, amino acids, and bicarbonate via coupled transcellular transport and bulk osmosis. Active secretion of organic anions and cations is mediated by polarized transporter proteins: organic anion transporters (OAT1/OAT3) on the basolateral membrane accept substrates from peritubular blood, while multidrug and toxin extrusion proteins (MATE1/MATE2) efflux them apically into the tubular lumen. Organic cation transporter 2 (OCT2) similarly orchestrates uptake of cationic drugs such as metformin and cisplatin. This transporter polarity, lost in immortalized 2D cultures but restored under microfluidic shear stress, is the central determinant of drug-induced nephrotoxicity risk and pharmacokinetic behavior. Functional readouts include glucose clearance, para-aminohippuric acid (PAH) secretion, creatinine handling, and transporter inhibition assays.

#### 3.3 Fluid Shear Stress and Mechanotransduction

Renal tubular epithelial cells are constitutively exposed to luminal fluid shear stress of  $\sim 0.1\text{--}1.0$  dyne/cm<sup>2</sup> arising from glomerular filtrate flow. This mechanical signal, sensed primarily by primary cilia projecting apically from PT cells, which activates polycystin-1 (PC1) and polycystin-2 (PC2) mechanoreceptors, triggering Ca<sup>2+</sup> influx and downstream kinase cascades that maintain epithelial polarity, tight junction integrity, and mitochondrial oxidative function. In static culture, PT cells lose apical brush border microvilli, primary cilia, and asymmetric transporter localization within days. Restoration of physiological shear stress in OoC recovers all these features and dramatically sensitizes cells to nephrotoxins at concentrations 20-fold lower than in static wells, a critical advantage for drug safety screening.

### 3.4 Epithelial Barrier Integrity

Paracellular permeability is regulated by tight junction (TJ) complexes, including occludin, claudins, and zonula occludens-1 (ZO-1) that form the primary seal between adjacent epithelial cells. Barrier integrity is measured quantitatively by transepithelial electrical resistance (TEER), expressed in  $\Omega\cdot\text{cm}^2$ , and by fluorescent tracer permeability (FITC-dextran, inulin-FITC, albumin-FITC). In the glomerular compartment, TJ-like slit diaphragm proteins (nephrin, podocin, CD2AP) fulfill the equivalent barrier function. Loss of barrier integrity manifested as TEER decline and albumin leakage, is the earliest functional marker of nephrotoxic injury and glomerular disease, making it the most widely used on-chip endpoint for drug screening and disease modeling.

## 4. Kidney-on-a-Chip Systems Published 2021–2025: Design, Aspects Described, and Key Findings

The following table summarizes five landmark kidney-on-chip publications from the past five years, cataloguing their chip design, cell sources, physiological aspects captured, and primary findings.

| Ref. | Year / Authors                                 | Chip Design                                                                                                                   | Aspects Described                                                                                                        | Key Findings                                                                                                                    | Limitations                                                                                                                   |
|------|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| [1]  | Ioannidis et al. 2022 (Dynamix platform)       | Vascularized PT-on-chip; PDMS microchannel; integrated metabolic microsensors (O <sub>2</sub> , glucose, lactate, glutamine)  | Real-time metabolic kinetics; shear-stress-driven polarization; gentamicin nephrotoxicity mechanism                      | Identified critical gentamicin concentration disrupting polarity 20× below TC50; SGLT2 inhibition rescued toxicity 10-fold      | No glomerular compartment; uses immortalized RPTEC line; limited immune context                                               |
| [2]  | Zhang & Mahler 2023 (Lab Chip)                 | Multi-layer nephron chip: glomerulus + PT co-culture separated by porous membrane; parallel microchannels                     | Integrated GFB + PT in one device; serum protein filtration; glucose reabsorption; creatinine secretion; AKI modelling   | First chip expressing VE-cadherin, nephrin, ZO-1 simultaneously; cisplatin/adriamycin AKI recapitulated with functional decline | Immortalized cell lines only; no immune cells; static GBM (no de novo matrix formation); limited long-term culture (< 7 days) |
| [3]  | Kroll et al. 2024 (Biofabrication / Lewis lab) | 3D bioprinted, perfusable vascularized kidney organoid-on-chip; embedded endothelialized microchannels within organoid matrix | Vascularization; organoid maturation under flow; tubular and glomerular segment co-development; VEGF-driven angiogenesis | Flow enhanced organoid vascularization and maturation; reduced off-target cardiac differentiation; improved podocyte markers    | Complex fabrication; limited throughput; no mechanical filtration pressure; immune-naïve; no distal tubule or collecting duct |

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| [4] | Hiratsuka et al. 2023 (Nature Comm.) | hiPSC-derived kidney organoid-on-chip (ARPKD model); microfluidic chip with cyclic mechanical stretch; cyst formation assay                            | Disease modeling (ARPKD); mechanosensing pathomechanisms; cyst growth under flow vs. static; drug discovery validation                                     | First OoC model of ARPKD revealing flow-dependent cystogenesis; tolvaptan efficacy confirmed; PKHD1 mechanosensing pathway elucidated                                             | Single disease model; no vascular endothelium in tubular cysts; limited to ductal epithelium; patient iPSC variability    |
| [5] | Kroll, Uzel & Lewis 2023 (PNAS)      | Immune-infiltrated vascularized kidney organoid-on-chip; cocirculation of PBMCs and T cell bispecific antibodies (TCBs) through 3D vascularized matrix | Immunotherapy safety testing; T cell activation/killing kinetics; off-tumor on-target toxicity (WT1 antigen); CD8 <sup>+</sup> T cell expansion under flow | Dose-dependent selective T-cell killing of WT1 <sup>+</sup> podocytes; model distinguished on-target from off-tumor effects; first immune-competent kidney OoC for TCB assessment | No full nephron integration; organoid maturity heterogeneity; PBMC donor variability; no hormonal or acid-base regulation |

## 5. Detailed Design Architectures of Published Kidney-on-Chip Systems

### 5.1 Proximal Tubule Chips with Integrated Metabolic Sensing [Ref. 1]

The DynamiX platform described by Ioannidis et al. (2022) represents the most metabolically instrumented PT-on-chip to date. Primary human proximal tubular epithelial cells (PTECs) and microvascular endothelial cells are seeded on opposing surfaces of a polyethylene terephthalate (PET) porous membrane within a PDMS microchannel. Microsensors for dissolved oxygen, glucose, lactate, and glutamine are embedded directly in the tubular lumen. Physiological perfusion (0.2 dyne/cm<sup>2</sup>) is maintained by a syringe pump. This design uniquely captures real-time metabolic flux demonstrated by showing that gentamicin toxicity reversed the tricarboxylic acid cycle flux, causing a 40-fold increase in lipid accumulation and 3.2-fold elevation in glucose uptake.

### 5.2 Integrated Glomerulus–Proximal Tubule Nephron Chip [Ref. 2]

Zhang and Mahler (2023) constructed the first multi-layer nephron chip coupling glomerular and proximal tubule compartments on a single device. The chip architecture uses three PDMS layers:

- (i) A top vascular channel seeded with glomerular endothelial cells.
- (ii) A central porous polycarbonate membrane functionalized with collagen IV and laminin to simulate GBM.
- (iii) A lower tubular channel seeded with immortalized podocytes (upper face) and proximal tubular cells (lower face). Parallel microchannels below the PT layer simulate peritubular capillary flow. This geometry enables sequential filtration (glomerulus) reabsorption/secretion (PT) in a single pass, providing the most nephron-complete architecture demonstrated with functional measurements of all three major transport events.

### 5.3 3D Bioprinted Vascularized Kidney Organoid-on-Chip [Ref. 3]

Kroll et al. (2024) from the Lewis laboratory at Harvard extended the flow-enhanced kidney organoid platform by embedding 3D bioprinted endothelialized channels directly within hiPSC-derived kidney organoid matrices. Organoids are cast in a fibrinogen/gelatin bioink, and open microchannels of defined geometry (~400 µm diameter) are printed through the organoid mass. Human umbilical vein endothelial cells (HUVECs) are then perfused through and coat the channels, creating a vascularized organoid construct. Continuous perfusion at 1–10 µL/min for 7–14 days drives angiogenic sprouting, reduces off-target cardiac and neural differentiation, and significantly upregulates

podocyte maturation markers (WT1, NPHS1, NPHS2) and proximal tubule identity genes (LRP2/megalin, CUBN, SLC34A1).

#### 5.4 Kidney Organoid-on-Chip for ARPKD Disease Modelling [Ref. 4]

Hiratsuka et al. (2023) adapted hiPSC-derived kidney organoids from patients with PKHD1-mutant ARPKD onto a PDMS microfluidic chip that incorporates cyclic mechanical stretch. The device contains a central culture chamber flanked by two vacuum chambers; applying alternating vacuum cyclically deforms the PDMS walls, imposing 10% linear stretch at 0.2 Hz on the embedded organoid, mimicking urinary flow-induced pressure. Critically, flow and stretch dramatically accelerated cystogenesis in ARPKD organoids relative to static culture, revealing that mechanosensing through the PKHD1-dependent pathway is a key driver of cyst expansion, a finding directly relevant to clinical observations of disease acceleration under urinary obstruction. Tolvaptan (a vasopressin V2R antagonist) significantly reduced cyst growth on-chip, thereby validating the model pharmacologically.

#### 5.5 Immune-Infiltrated Kidney Organoid-on-Chip for Immunotherapy Safety [Ref. 5]

The PNAS 2023 study from the Lewis laboratory created the first immune-competent kidney OoC for assessing T cell bispecific antibodies (TCBs). The 3D vascularized kidney organoid-on-chip (described in Ref. 3 architecture) was extended by cocirculating peripheral blood mononuclear cells (PBMCs) through the endothelialized channels alongside a WT1-targeted TCB. WT1 (Wilms Tumor 1 antigen) is a validated tumor antigen expressed both in leukemias and in healthy kidney podocytes, creating an on-target but off-tumor toxicity concern. The chip demonstrated rapid CD8<sup>+</sup> T cell expansion under flow, dose-dependent podocyte killing at TCB concentrations correlating with clinical dose-limiting toxicities, and immune exclusion at subtherapeutic concentrations establishing OoC as a platform for human-relevant immuno-oncology safety prediction unavailable in standard animal models.

## 6. Critical Unresolved Demerits Across Current Kidney-on-Chip Systems

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Despite remarkable advances, current kidney-on-chip platforms share several fundamental limitations that remain unsolved across all five reviewed systems:

### 6.1 Absence of Complete Nephron Architecture

No published chip integrates all five functionally distinct nephron segments: glomerulus, proximal convoluted tubule, loop of Henle, distal convoluted tubule, and collecting duct in their correct spatial sequence with appropriate cell types. The loop of Henle, which establishes the corticomedullary osmotic gradient critical for urine concentration, is entirely absent from every existing OoC design. Distal tubule aldosterone-regulated sodium reabsorption and collecting duct aquaporin-2-mediated water transport remain unmodeled. This omission means that no existing chip can reproduce net urine concentrating capacity a fundamental renal function nor can it accurately model diseases such as nephrogenic diabetes insipidus or hyperaldosteronism.

### 6.2 Immature or Immortalized Cell Sources

The majority of PT-on-chip studies use either immortalized human kidney cell lines (HK-2, RPTEC/TERT1) or early-passage primary PTECs, both of which inadequately express drug transporters (OAT1/3, OCT2, MATE1/2) at physiological levels and lack proper apical brush border morphology. While hiPSC-derived organoid cells improve genetic fidelity, their differentiation efficiency is variable between donors and protocols, their maturation state corresponds approximately to fetal kidney (second trimester), and they co-express off-target neural and cardiac markers. No chip system has yet employed fully adult-maturity kidney cells derived through directed differentiation or direct reprogramming that quantitatively match primary adult PTEC transporter expression.

### 6.3 Absence of Endocrine and Systemic Crosstalk

The kidney is an endocrine organ producing erythropoietin, renin, and 1,25-dihydroxyvitamin D<sub>3</sub>, while also responding to angiotensin II, aldosterone, antidiuretic hormone (ADH/vasopressin), parathyroid hormone, and atrial natriuretic peptide. None of the reviewed chips incorporate hormonal stimulation or feedback loops. Renin–angiotensin–aldosterone system (RAAS) signalling, which is central to hypertensive nephropathy, diabetic nephropathy progression, and cardiorenal syndrome, cannot be studied in any existing KoC without exogenous hormone supplementation that fails to capture systemic feedback dynamics.

### 6.4 No Immune-Stromal Integration in Disease-Relevant Contexts

While Ref. 5 introduced circulating T cells for immunotherapy testing, the complex resident immune environment of the kidney, including dendritic cells, macrophages (including tissue-resident kidney macrophages), mast cells, and NK cells, is absent from all chips. Renal fibrosis, the common endpoint of chronic kidney disease (CKD) regardless of etiology, requires myofibroblast activation from pericytes, tubular EMT, and macrophage-to-fibroblast crosstalk, none of which is captured. This limits the applicability of current KoC models to acute nephrotoxicity and early-stage disease, while the clinically most prevalent condition (CKD affecting >10% of the global population) remains essentially unmodeled.

### 6.5 Lack of Standardization, Scalability, and Validation Against Clinical Endpoints

Chip geometries, flow rates, cell seeding protocols, membrane materials, and functional readout methodologies differ substantially between laboratories, making inter-study comparison unreliable. PDMS, the near-universal chip material, nonspecifically adsorbs hydrophobic compounds ( $\log P > 1.5$ ), compromising dose-response measurements for the majority of drug candidates. No kidney OoC system has undergone formal regulatory qualification by the FDA or EMA as a validated testing tool, and none has been benchmarked against a sufficiently large clinical nephrotoxicity dataset to establish sensitivity, specificity, and predictive accuracy. High fabrication complexity and cost (~\$50–500 per chip) preclude the high-throughput screening applications that would most benefit drug development.

## 7. Extended Abstract

### Toward a Physiologically Complete Kidney-on-a-Chip: Advances, Unresolved Gaps, and the Path to Regulatory-Grade Microphysiological Renal Models

**Significance.** Kidney-on-a-chip platforms are emerging as the highest-fidelity in vitro surrogates for human renal physiology, capable of recapitulating glomerular filtration, proximal tubular drug transport, and nephrotoxic injury responses that static cell cultures fail to model. Their integration into regulatory frameworks as New Approach Methodologies promises to reduce the ~20% drug attrition rate attributable to renal toxicity. Yet critical architectural, cellular, and systemic gaps prevent existing chips from representing the full nephron. This work critically synthesizes the design landscape, identifies unresolved demerits, and proposes a roadmap toward a multi-segment, immune-competent, hormone-responsive kidney microphysiological system meeting FDA qualification criteria.

**Abstract.** The human kidney performs ~180 L of plasma filtration per day, reabsorbs essential solutes through polarized transporter arrays, and maintains systemic homeostasis through endocrine and paracrine signaling. Failure of these functions whether through drug-induced nephrotoxicity, autoimmune glomerulonephritis, diabetic nephropathy, or inherited ciliopathies, accounts for an enormous global disease burden, yet the translational gap between rodent models and human renal disease outcomes remains profound. Kidney-on-a-chip (KoC) systems have emerged over the past decade as microfluidic microphysiological platforms integrating living human kidney

cells within architectures that restore physiological shear stress, cell polarity, transporter activity, and tissue-level filtration; making them among the most clinically relevant New Approach Methodologies (NAMs) available.

Here, we provide a comprehensive critical synthesis of five landmark kidney-on-chip publications from 2022–2024, spanning the major design paradigms:

- (i) Metabolically instrumented proximal tubule chips with real-time sensing (Ioannidis et al., 2022)
- (ii) Integrated glomerulus–proximal tubule nephron chips for full filtration-to-secretion modeling (Zhang & Mahler, 2023)
- (iii) 3D bioprinted vascularized kidney organoid-on-chip for flow-enhanced maturation (Kroll et al., 2024)
- (iv) Mechanosensitive ARPKD organoid chips for disease modeling and drug validation (Hiratsuka et al., 2023)
- (v) Immune-infiltrated vascularized kidney organoid-on-chip for immunotherapy safety assessment (Kroll, Uzel & Lewis, 2023).

Collectively, these platforms advance the field by demonstrating the restoration of drug transporter polarity, GFB charge selectivity, cyst mechanopathology, and T cell-mediated immunotoxicity, all within human-cell-based microfluidic devices.

Despite this progress, we identify five critical and interrelated unresolved demerits. First, no system captures the complete nephron: the loop of Henle, distal convoluted tubule, and collecting duct and the osmotic gradient they generate are entirely absent, precluding modeling of urinary concentration, aldosterone responsiveness, and a broad class of renal disorders. Second, cell maturity remains sub-physiological: iPSC-derived cells correspond to fetal kidney developmental stages with transporter expression 10–100-fold below adult levels, while immortalized cell lines lack authentic brush border morphology and primary cilia. Third, endocrine integration is non-existent: RAAS signaling, ADH-mediated water transport, erythropoietin production, and calcitriol synthesis hallmarks of renal endocrine function cannot be modeled in any existing chip. Fourth, immune and stromal compartments are rudimentary: fibrosis, the universal endpoint of progressive CKD, requires myofibroblast-immune crosstalk unavailable in current designs. Fifth, no KoC system has been validated against clinical nephrotoxicity cohorts at a scale sufficient for regulatory qualification.

We propose that the next generation of kidney MPS must prioritize:

- (a) Multi-segment sequential nephron architecture incorporating loop of Henle epithelium derived from ureteric bud and metanephric mesenchyme lineages.
- (b) Direct reprogramming strategies or organoid maturation protocols achieving adult-equivalent transporter expression.
- (c) Integration of peritubular endocrine signaling circuits through recirculating media containing physiological hormone concentrations.
- (d) Inclusion of resident macrophages and myofibroblast precursors for fibrosis modelling
- (e) Adoption of thermoplastic chip materials (cyclic olefin copolymer, COC; polystyrene) to eliminate PDMS drug absorption and enable high-throughput manufacturing.

Achieving these milestones will position kidney-on-chip systems as fully validated NAM tools capable of supporting IND-enabling nephrotoxicity submissions to the FDA fundamentally transforming preclinical renal safety assessment and rare kidney disease drug development.

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